

Assignment: Backend Development with Laravel

Assignment Overview

You will build a simple RESTful API for managing **Products** in an inventory system. This assignment tests your ability to scaffold a Laravel project, connect to a database, and create a working CRUD API.

Submission Deadline

Due before Week 4 starts (by **21th December 2025, 11:59 PM**)

Scenario

Part A: Project Setup (25 marks)

A small business needs a simple system to manage their product inventory. Your task is to create the backend API that will allow:

- View all products
- View a single product
- Add a new product
- Update a product
- Delete a product

Deliverables:

- Screenshot of `.env` database configuration (hide passwords)
- Screenshot of successful `php artisan migrate` output

Part B: Model & Migration (25 marks)

Create a **Product** model with the following fields:

Field	Type	Rules
id	Auto-increment	Primary key
name	String (255)	Required
description	Text	Nullable
price	Decimal (10,2)	Required
stock	Integer	Default: 0
created_at	Timestamp	Auto-generated
updated_at	Timestamp	Auto-generated

Deliverables:

- **Product.php** Model file in app/Models folder
- Migration file in **database/migrations/**
- Screenshot showing migration ran successfully

Part C: Controller & Routes (25 marks)

Create a `ProductController` with these endpoints:

Method	Endpoint	Action	Description
GET	<code>/api/products</code>	index	List all products
GET	<code>/api/products/{id}</code>	show	Get single product
POST	<code>/api/products</code>	store	Create new product
PUT	<code>/api/products/{id}</code>	update	Update product
DELETE	<code>/api/products/{id}</code>	destroy	Delete product

Requirements:

- All responses return JSON
- Use correct HTTP status codes:
 - 200 for success
 - 201 for created
 - 404 for not found
 - 422 for validation errors
- Validate input for `store` and `update`
- Handle missing products with 404 response

Deliverables:

- `app/Http/Controllers/ProductController.php`
- `routes/api.php`

Part D: API Testing with Bruno (25 marks)

Test all 5 endpoints using Bruno and provide screenshots showing:

1. **GET `/api/products`** — Display empty product list (before creating any)
2. **POST `/api/products`** — Create a product (show 201 response)
3. **GET `/api/products`** — Display product list (after creating)
4. **GET `/api/products/{id}`** — Display single product
5. **PUT `/api/products/{id}`** — Update a product

6. **DELETE /api/products/{id}** — Delete a product

Deliverables:

- 6 screenshots clearly showing request and response in Bruno

Submission Checklist

Before submitting, ensure:

- Code pushed to GitHub
- README.md with setup instructions
- `.env.example` file included (no real passwords)
- Migrations run without errors
- All 5 endpoints working
- All screenshots in `screenshots/` folder

Submission Format

1. Create a GitHub repository, make sure the repo is set to 'public'.
2. Push all code
3. Add screenshots to `screenshots/` folder
4. Submit repository link to uzzair@invokeisdata.com